Filter Summary Report: CG,TIA,Automated,simple,Z1,Z4

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated simple Z1 Z4: $\frac{Z_1 Z_4 g_m}{2 Z_1 g_m + 2}$

$$H(z) = \frac{Z_1 Z_4 g_m}{2 Z_1 g_m + 2}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(R_1, \ \infty, \ \infty, \ \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \ \infty, \ \infty\right)$

$$H(s) = \frac{L_4 R_1 R_4 g_m s}{2 R_1 R_4 g_m + 2 R_4 + s^2 \left(2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_4\right) + s \left(2 L_4 R_1 g_m + 2 L_4\right)}$$

Parameters:

Q: $\frac{\sqrt{C_4} R_4}{\sqrt{L_4}}$
wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
bandwidth: $\frac{1}{C_4 R_4}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_4 g_m}{2 R_1 g_m + 2}$
Qz: None
Wz: None

3.2 BP-2 $Z(s) = \left(L_1 s, \ \infty, \ \infty, \ \frac{R_4}{C_4 R_4 s + 1}, \ \infty, \ \infty\right)$

$$H(s) = \frac{L_1 R_4 g_m s}{2 C_4 L_1 R_4 g_m s^2 + s \left(2 C_4 R_4 + 2 L_1 g_m\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{C_4} \sqrt{L_1} \sqrt{R_4} g_m \sqrt{\frac{1}{g_m}}}{C_4 R_4 + L_1 g_m}$
wo: $\frac{\sqrt{\frac{1}{g_m}}}{\sqrt{C_4} \sqrt{L_1} \sqrt{R_4}}$
bandwidth: $\frac{C_4 R_4 + L_1 g_m}{C_4 L_1 R_4 g_m}$
K-LP: 0
K-HP: 0
K-BP: $\frac{L_1 R_4 g_m}{2 C_4 R_4 + 2 L_1 g_m}$
Qz: None
Wz: None

3.3 BP-3 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \ \infty, \ \infty, \ R_4, \ \infty, \ \infty\right)$

$$H(s) = \frac{L_1 R_4 g_m s}{2 C_1 L_1 s^2 + 2 L_1 g_m s + 2}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1} g_m}$
wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_4}{2}$
Qz: None
Wz: None

3.4 BP-4 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1},\ \infty,\ \infty,\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 R_1 R_4 g_m s}{2C_1 L_1 R_1 s^2 + 2R_1 + s\left(2L_1 R_1 g_m + 2L_1\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{C_1 \sqrt{L_1} R_1}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
Qz: None
Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{R_4\left(C_4 L_4 s^2 + 1\right)}{C_4 L_4 s^2 + C_4 R_4 s + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_4 R_1 R_4 g_m s^2 + R_1 R_4 g_m}{2R_1 g_m + s^2\left(2C_4 L_4 R_1 g_m + 2C_4 L_4\right) + s\left(2C_4 R_1 R_4 g_m + 2C_4 R_4\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{L_4}}{\sqrt{C_4} R_4}$
wo: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$
bandwidth: $\frac{R_4}{L_4}$
K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
K-HP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$

4.2 BS-2 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \infty,\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 R_4 g_m s^2 + R_4 g_m}{2C_1 L_1 g_m s^2 + 2C_1 s + 2g_m}$$

Parameters:

Q: $\frac{\sqrt{L_1} g_m}{\sqrt{C_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{1}{L_1 g_m}$
K-LP: $\frac{R_4}{2}$
K-HP: $\frac{R_4}{2}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

4.3 BS-3 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \infty,\ R_4,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1L_1R_1R_4g_ms^2 + R_1R_4g_m}{2C_1R_1s + 2R_1g_m + s^2\left(2C_1L_1R_1g_m + 2C_1L_1\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{L_1}R_1g_m+\sqrt{L_1}}{\sqrt{C_1}R_1}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{R_1}{\sqrt{L_1}(\sqrt{L_1}R_1g_m+\sqrt{L_1})}$
K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
K-HP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

5 GE

5.1 GE-1 $Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ R_4,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1L_1R_4g_ms^2 + C_1R_1R_4g_ms + R_4g_m}{2C_1L_1g_ms^2 + 2g_m + s\left(2C_1R_1g_m + 2C_1\right)}$$

Parameters:

Q: $\frac{\sqrt{L_1}g_m}{\sqrt{C_1}R_1g_m+\sqrt{C_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{C_1}R_1g_m+\sqrt{C_1}}{\sqrt{C_1}L_1g_m}$
K-LP: $\frac{R_4}{2}$
K-HP: $\frac{R_4}{2}$
K-BP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
Qz: $\frac{\sqrt{L_1}}{\sqrt{C_1}R_1}$
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

5.2 GE-2 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \infty,\ R_4,\ \infty,\ \infty \right)$

$$H(s) = \frac{C_1L_1R_1R_4g_ms^2 + L_1R_4g_ms + R_1R_4g_m}{2L_1g_ms + 2R_1g_m + s^2\left(2C_1L_1R_1g_m + 2C_1L_1\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_1g_m+\sqrt{C_1}}{\sqrt{L_1}g_m}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{\sqrt{C_1}(\sqrt{C_1}R_1g_m+\sqrt{C_1})}$
K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
K-HP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
K-BP: $\frac{R_4}{2}$
Qz: $\frac{\sqrt{C_1}R_1}{\sqrt{L_1}}$
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_4 g_m}{2C_1 C_4 R_4 s^2 + 2g_m + s\left(2C_1 + 2C_4 R_4 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{g_m}}{C_1 + C_4 R_4 g_m}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}}$
bandwidth: $\frac{C_1 + C_4 R_4 g_m}{C_1 C_4 R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.2 LP-2 $Z(s) = \left(\frac{R_1}{C_1 R_1 s+1},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 R_4 g_m}{2C_1 C_4 R_1 R_4 s^2 + 2R_1 g_m + s\left(2C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_4\right) + 2}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1 g_m + 1}}{C_1 R_1 + C_4 R_1 R_4 g_m + C_4 R_4}$
wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}}$
bandwidth: $\frac{C_1 R_1 + C_4 R_1 R_4 g_m + C_4 R_4}{C_1 C_4 R_1 R_4}$
K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.3 LP-3 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 g_m}{2C_1 C_4 L_1 s^2 + 2C_4 L_1 g_m s + 2C_4}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1 g_m}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: $\frac{L_1 g_m}{2C_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.4 LP-4 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 R_1 g_m}{2C_1 C_4 L_1 R_1 s^2 + 2C_4 R_1 + s\left(2C_4 L_1 R_1 g_m + 2C_4 L_1\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}$

wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1}R_1g_m+\sqrt{L_1}}{C_1\sqrt{L_1}R_1}$
K-LP: $\frac{L_1g_m}{2C_4}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1R_1R_4g_ms + R_4g_m}{2g_m + s^2\left(2C_1C_4R_1R_4g_m + 2C_1C_4R_4\right) + s\left(2C_1R_1g_m + 2C_1 + 2C_4R_4g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}g_m\sqrt{\frac{g_m}{R_1g_m+1}}+\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{\frac{g_m}{R_1g_m+1}}}{C_1R_1g_m+C_1+C_4R_4g_m}$
wo: $\sqrt{\frac{g_m}{C_1C_4R_1R_4g_m+C_1C_4R_4}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{C_1C_4R_1R_4g_m+C_1C_4R_4}}(C_1R_1g_m+C_1+C_4R_4g_m)}{\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}g_m\sqrt{\frac{g_m}{R_1g_m+1}}+\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}\sqrt{\frac{g_m}{R_1g_m+1}}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_1R_1R_4g_m}{2C_1R_1g_m+2C_1+2C_4R_4g_m}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\frac{L_1 s}{C_1L_1s^2+1},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_1R_4g_ms + L_1g_m}{2C_1C_4L_1s^2 + 2C_4L_1g_ms + 2C_4}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1}g_m}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: $\frac{L_1g_m}{2C_4}$
K-HP: 0
K-BP: $\frac{R_4}{2}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\frac{L_1R_1 s}{C_1L_1R_1s^2+L_1s+R_1},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_1R_1R_4g_ms + L_1R_1g_m}{2C_1C_4L_1R_1s^2 + 2C_4R_1 + s\left(2C_4L_1R_1g_m + 2C_4L_1\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_1}{\sqrt{L_1}R_1g_m+\sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1}R_1g_m+\sqrt{L_1}}{C_1\sqrt{L_1}R_1}$
K-LP: $\frac{L_1g_m}{2C_4}$
K-HP: 0
K-BP: $\frac{R_1R_4g_m}{2R_1g_m+2}$

Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1,\ \infty,\ \infty,\ R_4,\ \infty,\ \infty)$

$$H(s) = \frac{R_1 R_4 g_m}{2R_1 g_m + 2}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 g_m}{s\left(2C_4 R_1 g_m + 2C_4\right)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1 R_4 g_m}{2R_1 g_m + s\left(2C_4 R_1 R_4 g_m + 2C_4 R_4\right) + 2}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1,\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 R_1 R_4 g_m s + R_1 g_m}{s\left(2C_4 R_1 g_m + 2C_4\right)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1,\ \infty,\ \infty,\ L_4 s + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_4 R_1 g_m s^2 + R_1 g_m}{s\left(2C_4 R_1 g_m + 2C_4\right)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{L_4 s}{C_4 L_4 s^2 + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_4 R_1 g_m s}{2R_1 g_m + s^2\left(2C_4 L_4 R_1 g_m + 2C_4 L_4\right) + 2}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1,\ \infty,\ \infty,\ L_4 s + R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_4 R_1 g_m s^2 + C_4 R_1 R_4 g_m s + R_1 g_m}{s\left(2C_4 R_1 g_m + 2C_4\right)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1,\ \infty,\ \infty,\ \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_4 R_1 R_4 g_m s^2 + L_4 R_1 g_m s + R_1 R_4 g_m}{2R_1 g_m + s^2\left(2C_4 L_4 R_1 g_m + 2C_4 L_4\right) + 2}$$

9.9 X-INVALID-ORDER-9 $Z(s) = (L_1 s,\ \infty,\ \infty,\ R_4,\ \infty,\ \infty)$

$$H(s) = \frac{L_1 R_4 g_m s}{2L_1 g_m s + 2}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(L_1 s,\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 g_m}{2C_4 L_1 g_m s + 2C_4}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(L_1 s,\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 R_4 g_m s + L_1 g_m}{2C_4 L_1 g_m s + 2C_4}$$

9.12 X-INVALID-ORDER-12 $Z(s) = \left(L_1 s,\ \infty,\ \infty,\ L_4 s + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 g_m s^2 + L_1 g_m}{2C_4 L_1 g_m s + 2C_4}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(L_1 s,\ \infty,\ \infty,\ \frac{L_4 s}{C_4 L_4 s^2 + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_4 g_m s^2}{2C_4 L_1 L_4 g_m s^3 + 2C_4 L_4 s^2 + 2L_1 g_m s + 2}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(L_1 s,\ \infty,\ \infty,\ L_4 s + R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 g_m s^2 + C_4 L_1 R_4 g_m s + L_1 g_m}{2C_4 L_1 g_m s + 2C_4}$$

9.15 X-INVALID-ORDER-15 $Z(s) = \left(L_1 s,\ \infty,\ \infty,\ \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_4 R_4 g_m s^2}{2C_4 L_1 L_4 R_4 g_m s^3 + 2R_4 + s^2\left(2C_4 L_4 R_4 + 2L_1 L_4 g_m\right) + s\left(2L_1 R_4 g_m + 2L_4\right)}$$

9.16 X-INVALID-ORDER-16 $Z(s) = \left(L_1 s,\ \infty,\ \infty,\ \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 R_4 g_m s^3 + L_1 L_4 g_m s^2 + L_1 R_4 g_m s}{2C_4 L_1 L_4 g_m s^3 + 2C_4 L_4 s^2 + 2L_1 g_m s + 2}$$

9.17 X-INVALID-ORDER-17 $Z(s) = \left(L_1 s,\ \infty,\ \infty,\ \frac{R_4\left(C_4 L_4 s^2 + 1\right)}{C_4 L_4 s^2 + C_4 R_4 s + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 R_4 g_m s^3 + L_1 R_4 g_m s}{2C_4 L_1 L_4 g_m s^3 + s^2\left(2C_4 L_1 R_4 g_m + 2C_4 L_4\right) + s\left(2C_4 R_4 + 2L_1 g_m\right) + 2}$$

9.18 X-INVALID-ORDER-18 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_4 g_m}{2C_1 s + 2g_m}$$

9.19 X-INVALID-ORDER-19 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{g_m}{2C_1C_4s^2 + 2C_4g_ms}$$

9.20 X-INVALID-ORDER-20 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4R_4g_ms + g_m}{2C_1C_4s^2 + 2C_4g_ms}$$

9.21 X-INVALID-ORDER-21 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ L_4s + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_4g_ms^2 + g_m}{2C_1C_4s^2 + 2C_4g_ms}$$

9.22 X-INVALID-ORDER-22 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{L_4s}{C_4L_4s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_4g_ms}{2C_1C_4L_4s^3 + 2C_1s + 2C_4L_4g_ms^2 + 2g_m}$$

9.23 X-INVALID-ORDER-23 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ L_4s + R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_4g_ms^2 + C_4R_4g_ms + g_m}{2C_1C_4s^2 + 2C_4g_ms}$$

9.24 X-INVALID-ORDER-24 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_4R_4g_ms}{2C_1C_4L_4R_4s^3 + 2R_4g_m + s^2\left(2C_1L_4 + 2C_4L_4R_4g_m\right) + s\left(2C_1R_4 + 2L_4g_m\right)}$$

9.25 X-INVALID-ORDER-25 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_4R_4g_ms^2 + L_4g_ms + R_4g_m}{2C_1C_4L_4s^3 + 2C_1s + 2C_4L_4g_ms^2 + 2g_m}$$

9.26 X-INVALID-ORDER-26 $Z(s) = \left(\frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4\left(C_4L_4s^2+1\right)}{C_4L_4s^2+C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_4R_4g_ms^2 + R_4g_m}{2C_1C_4L_4s^3 + 2g_m + s^2\left(2C_1C_4R_4 + 2C_4L_4g_m\right) + s\left(2C_1 + 2C_4R_4g_m\right)}$$

9.27 X-INVALID-ORDER-27 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1R_4g_m}{2C_1R_1s + 2R_1g_m + 2}$$

9.28 X-INVALID-ORDER-28 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{R_1g_m}{2C_1C_4R_1s^2 + s\left(2C_4R_1g_m + 2C_4\right)}$$

9.29 X-INVALID-ORDER-29 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4R_1R_4g_ms + R_1g_m}{2C_1C_4R_1s^2 + s\left(2C_4R_1g_m + 2C_4\right)}$$

9.30 X-INVALID-ORDER-30 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ L_4s + \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_4R_1g_ms^2 + R_1g_m}{2C_1C_4R_1s^2 + s\left(2C_4R_1g_m + 2C_4\right)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ \frac{L_4s}{C_4L_4s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_4R_1g_ms}{2C_1C_4L_4R_1s^3 + 2C_1R_1s + 2R_1g_m + s^2\left(2C_4L_4R_1g_m + 2C_4L_4\right) + 2}$$

9.32 X-INVALID-ORDER-32 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ L_4s + R_4 + \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_4R_1g_ms^2 + C_4R_1R_4g_ms + R_1g_m}{2C_1C_4R_1s^2 + s\left(2C_4R_1g_m + 2C_4\right)}$$

9.33 X-INVALID-ORDER-33 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_4R_1R_4g_ms}{2C_1C_4L_4R_1R_4s^3 + 2R_1R_4g_m + 2R_4 + s^2\left(2C_1L_4R_1 + 2C_4L_4R_1R_4g_m + 2C_4L_4R_4\right) + s\left(2C_1R_1R_4 + 2L_4R_1g_m + 2L_4\right)}$$

9.34 X-INVALID-ORDER-34 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_4R_1R_4g_ms^2 + L_4R_1g_ms + R_1R_4g_m}{2C_1C_4L_4R_1s^3 + 2C_1R_1s + 2R_1g_m + s^2\left(2C_4L_4R_1g_m + 2C_4L_4\right) + 2}$$

9.35 X-INVALID-ORDER-35 $Z(s) = \left(\frac{R_1}{C_1R_1s+1},\ \infty,\ \infty,\ \frac{R_4\left(C_4L_4s^2+1\right)}{C_4L_4s^2+C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_4R_1R_4g_ms^2 + R_1R_4g_m}{2C_1C_4L_4R_1s^3 + 2R_1g_m + s^2\left(2C_1C_4R_1R_4 + 2C_4L_4R_1g_m + 2C_4L_4\right) + s\left(2C_1R_1 + 2C_4R_1R_4g_m + 2C_4R_4\right) + 2}$$

9.36 X-INVALID-ORDER-36 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ R_4,\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1R_1R_4g_ms + R_4g_m}{2g_m + s\left(2C_1R_1g_m + 2C_1\right)}$$

9.37 X-INVALID-ORDER-37 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1R_1g_ms + g_m}{2C_4g_ms + s^2\left(2C_1C_4R_1g_m + 2C_1C_4\right)}$$

9.38 X-INVALID-ORDER-38 $Z(s) = \left(R_1 + \frac{1}{C_1s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4R_1R_4g_ms^2 + g_m + s\left(C_1R_1g_m + C_4R_4g_m\right)}{2C_4g_ms + s^2\left(2C_1C_4R_1g_m + 2C_1C_4\right)}$$

9.39 X-INVALID-ORDER-39 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ L_4 s + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_4R_1g_ms^3 + C_1R_1g_ms + C_4L_4g_ms^2 + g_m}{2C_4g_ms + s^2\left(2C_1C_4R_1g_m + 2C_1C_4\right)}$$

9.40 X-INVALID-ORDER-40 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{L_4 s}{C_4L_4s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_4R_1g_ms^2 + L_4g_ms}{2C_4L_4g_ms^2 + 2g_m + s^3\left(2C_1C_4L_4R_1g_m + 2C_1C_4L_4\right) + s\left(2C_1R_1g_m + 2C_1\right)}$$

9.41 X-INVALID-ORDER-41 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ L_4 s + R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_4R_1g_ms^3 + g_m + s^2\left(C_1C_4R_1R_4g_m + C_4L_4g_m\right) + s\left(C_1R_1g_m + C_4R_4g_m\right)}{2C_4g_ms + s^2\left(2C_1C_4R_1g_m + 2C_1C_4\right)}$$

9.42 X-INVALID-ORDER-42 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_4R_1R_4g_ms^2 + L_4R_4g_ms}{2R_4g_m + s^3\left(2C_1C_4L_4R_1R_4g_m + 2C_1C_4L_4R_4\right) + s^2\left(2C_1L_4R_1g_m + 2C_1L_4 + 2C_4L_4R_4g_m\right) + s\left(2C_1R_1R_4g_m + 2C_1R_4 + 2L_4g_m\right)}$$

9.43 X-INVALID-ORDER-43 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_4R_1R_4g_ms^3 + R_4g_m + s^2\left(C_1L_4R_1g_m + C_4L_4R_4g_m\right) + s\left(C_1R_1R_4g_m + L_4g_m\right)}{2C_4L_4g_ms^2 + 2g_m + s^3\left(2C_1C_4L_4R_1g_m + 2C_1C_4L_4\right) + s\left(2C_1R_1g_m + 2C_1\right)}$$

9.44 X-INVALID-ORDER-44 $Z(s) = \left(R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4\left(C_4L_4s^2+1\right)}{C_4L_4s^2+C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_4R_1R_4g_ms^3 + C_1R_1R_4g_ms + C_4L_4R_4g_ms^2 + R_4g_m}{2g_m + s^3\left(2C_1C_4L_4R_1g_m + 2C_1C_4L_4\right) + s^2\left(2C_1C_4R_1R_4g_m + 2C_1C_4R_4 + 2C_4L_4g_m\right) + s\left(2C_1R_1g_m + 2C_1 + 2C_4R_4g_m\right)}$$

9.45 X-INVALID-ORDER-45 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1g_ms^2 + g_m}{2C_1C_4L_1g_ms^3 + 2C_1C_4s^2 + 2C_4g_ms}$$

9.46 X-INVALID-ORDER-46 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1R_4g_ms^2 + R_4g_m}{2C_1C_4L_1R_4g_ms^3 + 2g_m + s^2\left(2C_1C_4R_4 + 2C_1L_1g_m\right) + s\left(2C_1 + 2C_4R_4g_m\right)}$$

9.47 X-INVALID-ORDER-47 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_1R_4g_ms^3 + C_1L_1g_ms^2 + C_4R_4g_ms + g_m}{2C_1C_4L_1g_ms^3 + 2C_1C_4s^2 + 2C_4g_ms}$$

9.48 X-INVALID-ORDER-48 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \infty,\ L_4 s + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 g_m s^4 + g_m + s^2 \left(C_1 L_1 g_m + C_4 L_4 g_m\right)}{2 C_1 C_4 L_1 g_m s^3 + 2 C_1 C_4 s^2 + 2 C_4 g_m s}$$

9.49 X-INVALID-ORDER-49 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{L_4 s}{C_4 L_4 s^2 + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 L_4 g_m s^3 + L_4 g_m s}{2 C_1 C_4 L_1 L_4 g_m s^4 + 2 C_1 C_4 L_4 s^3 + 2 C_1 s + 2 g_m + s^2 \left(2 C_1 L_1 g_m + 2 C_4 L_4 g_m\right)}$$

9.50 X-INVALID-ORDER-50 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \infty,\ L_4 s + R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 g_m s^4 + C_1 C_4 L_1 R_4 g_m s^3 + C_4 R_4 g_m s + g_m + s^2 \left(C_1 L_1 g_m + C_4 L_4 g_m\right)}{2 C_1 C_4 L_1 g_m s^3 + 2 C_1 C_4 s^2 + 2 C_4 g_m s}$$

9.51 X-INVALID-ORDER-51 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 L_4 R_4 g_m s^3 + L_4 R_4 g_m s}{2 C_1 C_4 L_1 L_4 R_4 g_m s^4 + 2 R_4 g_m + s^3 \left(2 C_1 C_4 L_4 R_4 + 2 C_1 L_1 L_4 g_m\right) + s^2 \left(2 C_1 L_1 R_4 g_m + 2 C_1 L_4 + 2 C_4 L_4 R_4 g_m\right) + s \left(2 C_1 R_4 + 2 L_4 g_m\right)}$$

9.52 X-INVALID-ORDER-52 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_4 g_m s^4 + C_1 L_1 L_4 g_m s^3 + L_4 g_m s + R_4 g_m + s^2 \left(C_1 L_1 R_4 g_m + C_4 L_4 R_4 g_m\right)}{2 C_1 C_4 L_1 L_4 g_m s^4 + 2 C_1 C_4 L_4 s^3 + 2 C_1 s + 2 g_m + s^2 \left(2 C_1 L_1 g_m + 2 C_4 L_4 g_m\right)}$$

9.53 X-INVALID-ORDER-53 $Z(s) = \left(L_1 s + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4 \left(C_4 L_4 s^2 + 1\right)}{C_4 L_4 s^2 + C_4 R_4 s + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_4 g_m s^4 + R_4 g_m + s^2 \left(C_1 L_1 R_4 g_m + C_4 L_4 R_4 g_m\right)}{2 C_1 C_4 L_1 L_4 g_m s^4 + 2 g_m + s^3 \left(2 C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_4\right) + s^2 \left(2 C_1 C_4 R_4 + 2 C_1 L_1 g_m + 2 C_4 L_4 g_m\right) + s \left(2 C_1 + 2 C_4 R_4 g_m\right)}$$

9.54 X-INVALID-ORDER-54 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 R_4 g_m s}{2 C_1 C_4 L_1 R_4 s^3 + s^2 \left(2 C_1 L_1 + 2 C_4 L_1 R_4 g_m\right) + s \left(2 C_4 R_4 + 2 L_1 g_m\right) + 2}$$

9.55 X-INVALID-ORDER-55 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \infty,\ \frac{L_4 s}{C_4 L_4 s^2 + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_4 g_m s^2}{2 C_1 C_4 L_1 L_4 s^4 + 2 C_4 L_1 L_4 g_m s^3 + 2 L_1 g_m s + s^2 \left(2 C_1 L_1 + 2 C_4 L_4\right) + 2}$$

9.56 X-INVALID-ORDER-56 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \infty,\ \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_4 R_4 g_m s^2}{2 C_1 C_4 L_1 L_4 R_4 s^4 + 2 R_4 + s^3 \left(2 C_1 L_1 L_4 + 2 C_4 L_1 L_4 R_4 g_m\right) + s^2 \left(2 C_1 L_1 R_4 + 2 C_4 L_4 R_4 + 2 L_1 L_4 g_m\right) + s \left(2 L_1 R_4 g_m + 2 L_4\right)}$$

9.57 X-INVALID-ORDER-57 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1},\ \infty,\ \infty,\ \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 R_4 g_m s^3 + L_1 L_4 g_m s^2 + L_1 R_4 g_m s}{2 C_1 C_4 L_1 L_4 s^4 + 2 C_4 L_1 L_4 g_m s^3 + 2 L_1 g_m s + s^2 \left(2 C_1 L_1 + 2 C_4 L_4\right) + 2}$$

9.58 X-INVALID-ORDER-58 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \infty,\ \frac{R_4\left(C_4 L_4 s^2+1\right)}{C_4 L_4 s^2+C_4 R_4 s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 R_4 g_m s^3 + L_1 R_4 g_m s}{2C_1 C_4 L_1 L_4 s^4 + s^3 \left(2C_1 C_4 L_1 R_4 + 2C_4 L_1 L_4 g_m\right) + s^2 \left(2C_1 L_1 + 2C_4 L_1 R_4 g_m + 2C_4 L_4\right) + s \left(2C_4 R_4 + 2L_1 g_m\right) + 2}$$

9.59 X-INVALID-ORDER-59 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 g_m s^2 + C_1 R_1 g_m s + g_m}{2C_1 C_4 L_1 g_m s^3 + 2C_4 g_m s + s^2 \left(2C_1 C_4 R_1 g_m + 2C_1 C_4\right)}$$

9.60 X-INVALID-ORDER-60 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 R_4 g_m s^2 + C_1 R_1 R_4 g_m s + R_4 g_m}{2C_1 C_4 L_1 R_4 g_m s^3 + 2g_m + s^2 \left(2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_1 L_1 g_m\right) + s \left(2C_1 R_1 g_m + 2C_1 + 2C_4 R_4 g_m\right)}$$

9.61 X-INVALID-ORDER-61 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_4 L_1 R_4 g_m s^3 + g_m + s^2 \left(C_1 C_4 R_1 R_4 g_m + C_1 L_1 g_m\right) + s \left(C_1 R_1 g_m + C_4 R_4 g_m\right)}{2C_1 C_4 L_1 g_m s^3 + 2C_4 g_m s + s^2 \left(2C_1 C_4 R_1 g_m + 2C_1 C_4\right)}$$

9.62 X-INVALID-ORDER-62 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ L_4 s + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 g_m s^4 + C_1 C_4 L_4 R_1 g_m s^3 + C_1 R_1 g_m s + g_m + s^2 \left(C_1 L_1 g_m + C_4 L_4 g_m\right)}{2C_1 C_4 L_1 g_m s^3 + 2C_4 g_m s + s^2 \left(2C_1 C_4 R_1 g_m + 2C_1 C_4\right)}$$

9.63 X-INVALID-ORDER-63 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{L_4 s}{C_4 L_4 s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 L_4 g_m s^3 + C_1 L_4 R_1 g_m s^2 + L_4 g_m s}{2C_1 C_4 L_1 L_4 g_m s^4 + 2g_m + s^3 \left(2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4\right) + s^2 \left(2C_1 L_1 g_m + 2C_4 L_4 g_m\right) + s \left(2C_1 R_1 g_m + 2C_1\right)}$$

9.64 X-INVALID-ORDER-64 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ L_4 s + R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 g_m s^4 + g_m + s^3 \left(C_1 C_4 L_1 R_4 g_m + C_1 C_4 L_4 R_1 g_m\right) + s^2 \left(C_1 C_4 R_1 R_4 g_m + C_1 L_1 g_m + C_4 L_4 g_m\right) + s \left(C_1 R_1 g_m + C_4 R_4 g_m\right)}{2C_1 C_4 L_1 g_m s^3 + 2C_4 g_m s + s^2 \left(2C_1 C_4 R_1 g_m + 2C_1 C_4\right)}$$

9.65 X-INVALID-ORDER-65 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2+L_4 s+R_4},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 L_4 R_4 g_m s^3 + C_1 L_4 R_1 R_4 g_m s^2 + L_4 R_4 g_m s}{2C_1 C_4 L_1 L_4 R_4 g_m s^4 + 2R_4 g_m + s^3 \left(2C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_4 + 2C_1 L_1 L_4 g_m\right) + s^2 \left(2C_1 L_1 R_4 g_m + 2C_1 L_4 R_1 g_m + 2C_1 L_4 + 2C_4 L_4 R_4 g_m\right) + s \left(2C_1 R_1 R_4 g_m + 2C_1 R_4 + 2L_4 g_m\right)}$$

9.66 X-INVALID-ORDER-66 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{C_4 L_4 R_4 s^2+L_4 s+R_4}{C_4 L_4 s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_4 g_m s^4 + R_4 g_m + s^3 \left(C_1 C_4 L_4 R_1 R_4 g_m + C_1 L_1 L_4 g_m\right) + s^2 \left(C_1 L_1 R_4 g_m + C_1 L_4 R_1 g_m + C_4 L_4 R_4 g_m\right) + s \left(C_1 R_1 R_4 g_m + L_4 g_m\right)}{2C_1 C_4 L_1 L_4 g_m s^4 + 2g_m + s^3 \left(2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4\right) + s^2 \left(2C_1 L_1 g_m + 2C_4 L_4 g_m\right) + s \left(2C_1 R_1 g_m + 2C_1\right)}$$

9.67 X-INVALID-ORDER-67

$Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s},\ \infty,\ \infty,\ \frac{R_4\left(C_4 L_4 s^2+1\right)}{C_4 L_4 s^2 + C_4 R_4 s + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_4 g_m s^4 + C_1 C_4 L_4 R_1 R_4 g_m s^3 + C_1 R_1 R_4 g_m s + R_4 g_m + s^2 \left(C_1 L_1 R_4 g_m + C_4 L_4 R_4 g_m\right)}{2C_1 C_4 L_1 L_4 g_m s^4 + 2g_m + s^3 \left(2C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4\right) + s^2 \left(2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_1 L_1 g_m + 2C_4 L_4 g_m\right) + s \left(2C_1 R_1 g_m + 2C_1 + 2C_4 R_4 g_m\right)}$$

9.68 X-INVALID-ORDER-68

$Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 R_1 R_4 g_m s}{2C_1 C_4 L_1 R_1 R_4 s^3 + 2R_1 + s^2 \left(2C_1 L_1 R_1 + 2C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_4\right) + s \left(2C_4 R_1 R_4 + 2L_1 R_1 g_m + 2L_1\right)}$$

9.69 X-INVALID-ORDER-69

$Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1},\ \infty,\ \infty,\ \frac{L_4 s}{C_4 L_4 s^2 + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_4 R_1 g_m s^2}{2C_1 C_4 L_1 L_4 R_1 s^4 + 2R_1 + s^3 \left(2C_4 L_1 L_4 R_1 g_m + 2C_4 L_1 L_4\right) + s^2 \left(2C_1 L_1 R_1 + 2C_4 L_4 R_1\right) + s \left(2L_1 R_1 g_m + 2L_1\right)}$$

9.70 X-INVALID-ORDER-70

$Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1},\ \infty,\ \infty,\ \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4},\ \infty,\ \infty\right)$

$$H(s) = \frac{L_1 L_4 R_1 R_4 g_m s^2}{2C_1 C_4 L_1 L_4 R_1 R_4 s^4 + 2R_1 R_4 + s^3 \left(2C_1 L_1 L_4 R_1 + 2C_4 L_1 L_4 R_1 R_4 g_m + 2C_4 L_1 L_4 R_4\right) + s^2 \left(2C_1 L_1 R_1 R_4 + 2C_4 L_4 R_1 R_4 + 2L_1 L_4 R_1 g_m + 2L_1 L_4\right) + s \left(2L_1 R_1 R_4 g_m + 2L_1 R_4 + 2L_4 R_1\right)}$$

9.71 X-INVALID-ORDER-71

$Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1},\ \infty,\ \infty,\ \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_4 g_m s^3 + L_1 L_4 R_1 g_m s^2 + L_1 R_1 R_4 g_m s}{2C_1 C_4 L_1 L_4 R_1 s^4 + 2R_1 + s^3 \left(2C_4 L_1 L_4 R_1 g_m + 2C_4 L_1 L_4\right) + s^2 \left(2C_1 L_1 R_1 + 2C_4 L_4 R_1\right) + s \left(2L_1 R_1 g_m + 2L_1\right)}$$

9.72 X-INVALID-ORDER-72

$Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1},\ \infty,\ \infty,\ \frac{R_4\left(C_4 L_4 s^2+1\right)}{C_4 L_4 s^2 + C_4 R_4 s + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_4 g_m s^3 + L_1 R_1 R_4 g_m s}{2C_1 C_4 L_1 L_4 R_1 s^4 + 2R_1 + s^3 \left(2C_1 C_4 L_1 R_1 R_4 + 2C_4 L_1 L_4 R_1 g_m + 2C_4 L_1 L_4\right) + s^2 \left(2C_1 L_1 R_1 + 2C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_4 + 2C_4 L_4 R_1\right) + s \left(2C_4 R_1 R_4 + 2L_1 R_1 g_m + 2L_1\right)}$$

9.73 X-INVALID-ORDER-73

$Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1},\ \infty,\ \infty,\ \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 R_1 g_m s^2 + L_1 g_m s + R_1 g_m}{2C_4 L_1 g_m s^2 + s^3 \left(2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1\right) + s \left(2C_4 R_1 g_m + 2C_4\right)}$$

9.74 X-INVALID-ORDER-74

$Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1},\ \infty,\ \infty,\ \frac{R_4}{C_4 R_4 s + 1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 L_1 R_1 R_4 g_m s^2 + L_1 R_4 g_m s + R_1 R_4 g_m}{2R_1 g_m + s^3 \left(2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_4\right) + s^2 \left(2C_1 L_1 R_1 g_m + 2C_1 L_1 + 2C_4 L_1 R_4 g_m\right) + s \left(2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2L_1 g_m\right) + 2}$$

9.75 X-INVALID-ORDER-75

$Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1},\ \infty,\ \infty,\ R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1 C_4 L_1 R_1 R_4 g_m s^3 + R_1 g_m + s^2 \left(C_1 L_1 R_1 g_m + C_4 L_1 R_4 g_m\right) + s \left(C_4 R_1 R_4 g_m + L_1 g_m\right)}{2C_4 L_1 g_m s^2 + s^3 \left(2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1\right) + s \left(2C_4 R_1 g_m + 2C_4\right)}$$

9.76 X-INVALID-ORDER-76 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \infty,\ L_4s+\frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_1L_4R_1g_ms^4 + C_4L_1L_4g_ms^3 + L_1g_ms + R_1g_m + s^2\left(C_1L_1R_1g_m + C_4L_4R_1g_m\right)}{2C_4L_1g_ms^2 + s^3\left(2C_1C_4L_1R_1g_m + 2C_1C_4L_1\right) + s\left(2C_4R_1g_m + 2C_4\right)}$$

9.77 X-INVALID-ORDER-77 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \infty,\ \frac{L_4s}{C_4L_4s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1L_4R_1g_ms^3 + L_1L_4g_ms^2 + L_4R_1g_ms}{2C_4L_1L_4g_ms^3 + 2L_1g_ms + 2R_1g_m + s^4\left(2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4\right) + s^2\left(2C_1L_1R_1g_m + 2C_1L_1 + 2C_4L_4R_1g_m + 2C_4L_4\right) + 2}$$

9.78 X-INVALID-ORDER-78 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \infty,\ L_4s+R_4+\frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_1L_4R_1g_ms^4 + R_1g_m + s^3\left(C_1C_4L_1R_1R_4g_m + C_4L_1L_4g_m\right) + s^2\left(C_1L_1R_1g_m + C_4L_1R_4g_m + C_4L_4R_1g_m\right) + s\left(C_4R_1R_4g_m + L_1g_m\right)}{2C_4L_1g_ms^2 + s^3\left(2C_1C_4L_1R_1g_m + 2C_1C_4L_1\right) + s\left(2C_4R_1g_m + 2C_4\right)}$$

9.79 X-INVALID-ORDER-79 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \infty,\ \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1L_4R_1R_4g_ms^3 + L_1L_4R_4g_ms^2 + L_4R_1R_4g_ms}{2R_1R_4g_m + 2R_4 + s^4\left(2C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_4\right) + s^3\left(2C_1L_1L_4R_1g_m + 2C_1L_1L_4 + 2C_4L_1L_4R_4g_m\right) + s^2\left(2C_1L_1R_1R_4g_m + 2C_1L_1R_4 + 2C_4L_4R_1R_4g_m + 2C_4L_4R_4 + 2L_1L_4g_m\right) + s\left(2L_1R_4g_m + 2L_4R_1g_m + 2L_4\right)}$$

9.80 X-INVALID-ORDER-80 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \infty,\ \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_1L_4R_1R_4g_ms^4 + R_1R_4g_m + s^3\left(C_1L_1L_4R_1g_m + C_4L_1L_4R_4g_m\right) + s^2\left(C_1L_1R_1R_4g_m + C_4L_4R_1R_4g_m + L_1L_4g_m\right) + s\left(L_1R_4g_m + L_4R_1g_m\right)}{2C_4L_1L_4g_ms^3 + 2L_1g_ms + 2R_1g_m + s^4\left(2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4\right) + s^2\left(2C_1L_1R_1g_m + 2C_1L_1 + 2C_4L_4R_1g_m + 2C_4L_4\right) + 2}$$

9.81 X-INVALID-ORDER-81 $Z(s) = \left(\frac{C_1L_1R_1s^2+L_1s+R_1}{C_1L_1s^2+1},\ \infty,\ \infty,\ \frac{R_4\left(C_4L_4s^2+1\right)}{C_4L_4s^2+C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_1L_4R_1R_4g_ms^4 + C_4L_1L_4R_4g_ms^3 + L_1R_4g_ms + R_1R_4g_m + s^2\left(C_1L_1R_1R_4g_m + C_4L_4R_1R_4g_m\right)}{2R_1g_m + s^4\left(2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4\right) + s^3\left(2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_4 + 2C_4L_1L_4g_m\right) + s^2\left(2C_1L_1R_1g_m + 2C_1L_1 + 2C_4L_1R_4g_m + 2C_4L_4R_1g_m + 2C_4L_4\right) + s\left(2C_4R_1R_4g_m + 2C_4R_4 + 2L_1g_m\right) + 2}$$

9.82 X-INVALID-ORDER-82 $Z(s) = \left(\frac{R_1\left(C_1L_1s^2+1\right)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \infty,\ \frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1R_1g_ms^2 + R_1g_m}{2C_1C_4R_1s^2 + s^3\left(2C_1C_4L_1R_1g_m + 2C_1C_4L_1\right) + s\left(2C_4R_1g_m + 2C_4\right)}$$

9.83 X-INVALID-ORDER-83 $Z(s) = \left(\frac{R_1\left(C_1L_1s^2+1\right)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \infty,\ \frac{R_4}{C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1R_1R_4g_ms^2 + R_1R_4g_m}{2R_1g_m + s^3\left(2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_4\right) + s^2\left(2C_1C_4R_1R_4 + 2C_1L_1R_1g_m + 2C_1L_1\right) + s\left(2C_1R_1 + 2C_4R_1R_4g_m + 2C_4R_4\right) + 2}$$

9.84 X-INVALID-ORDER-84 $Z(s) = \left(\frac{R_1\left(C_1L_1s^2+1\right)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \infty,\ R_4+\frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_1R_1R_4g_ms^3 + C_1L_1R_1g_ms^2 + C_4R_1R_4g_ms + R_1g_m}{2C_1C_4R_1s^2 + s^3\left(2C_1C_4L_1R_1g_m + 2C_1C_4L_1\right) + s\left(2C_4R_1g_m + 2C_4\right)}$$

9.85 X-INVALID-ORDER-85 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \infty,\ L_4s+\frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_1L_4R_1g_ms^4 + R_1g_m + s^2\left(C_1L_1R_1g_m + C_4L_4R_1g_m\right)}{2C_1C_4R_1s^2 + s^3\left(2C_1C_4L_1R_1g_m + 2C_1C_4L_1\right) + s\left(2C_4R_1g_m + 2C_4\right)}$$

9.86 X-INVALID-ORDER-86 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \infty,\ \frac{L_4s}{C_4L_4s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1L_4R_1g_ms^3 + L_4R_1g_ms}{2C_1C_4L_4R_1s^3 + 2C_1R_1s + 2R_1g_m + s^4\left(2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4\right) + s^2\left(2C_1L_1R_1g_m + 2C_1L_1 + 2C_4L_4R_1g_m + 2C_4L_4\right) + 2}$$

9.87 X-INVALID-ORDER-87 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \infty,\ L_4s+R_4+\frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_1L_4R_1g_ms^4 + C_1C_4L_1R_1R_4g_ms^3 + C_4R_1R_4g_ms + R_1g_m + s^2\left(C_1L_1R_1g_m + C_4L_4R_1g_m\right)}{2C_1C_4R_1s^2 + s^3\left(2C_1C_4L_1R_1g_m + 2C_1C_4L_1\right) + s\left(2C_4R_1g_m + 2C_4\right)}$$

9.88 X-INVALID-ORDER-88 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \infty,\ \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1L_1L_4R_1R_4g_ms^3 + L_4R_1R_4g_ms}{2R_1R_4g_m + 2R_4 + s^4\left(2C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_4\right) + s^3\left(2C_1C_4L_4R_1R_4 + 2C_1L_1L_4R_1g_m + 2C_1L_1L_4\right) + s^2\left(2C_1L_1R_1R_4g_m + 2C_1L_1R_4 + 2C_1L_4R_1 + 2C_4L_4R_1R_4g_m + 2C_4L_4R_4\right) + s\left(2C_1R_1R_4 + 2L_4R_1g_m + 2L_4\right)}$$

9.89 X-INVALID-ORDER-89 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \infty,\ \frac{C_4L_4R_4s^2+L_4s+R_4}{C_4L_4s^2+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_1L_4R_1R_4g_ms^4 + C_1L_1L_4R_1g_ms^3 + L_4R_1g_ms + R_1R_4g_m + s^2\left(C_1L_1R_1R_4g_m + C_4L_4R_1R_4g_m\right)}{2C_1C_4L_4R_1s^3 + 2C_1R_1s + 2R_1g_m + s^4\left(2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4\right) + s^2\left(2C_1L_1R_1g_m + 2C_1L_1 + 2C_4L_4R_1g_m + 2C_4L_4\right) + 2}$$

9.90 X-INVALID-ORDER-90 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1},\ \infty,\ \infty,\ \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_1C_4L_1L_4R_1R_4g_ms^4 + R_1R_4g_m + s^2\left(C_1L_1R_1R_4g_m + C_4L_4R_1R_4g_m\right)}{2R_1g_m + s^4\left(2C_1C_4L_1L_4R_1g_m + 2C_1C_4L_1L_4\right) + s^3\left(2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_4 + 2C_1C_4L_4R_1\right) + s^2\left(2C_1C_4R_1R_4 + 2C_1L_1R_1g_m + 2C_1L_1 + 2C_4L_4R_1g_m + 2C_4L_4\right) + s\left(2C_1R_1 + 2C_4R_1R_4g_m + 2C_4R_4\right) + 2}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\frac{L_1s}{C_1L_1s^2+1},\ \infty,\ \infty,\ L_4s+\frac{1}{C_4s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4L_1L_4g_ms^2 + L_1g_m}{2C_1C_4L_1s^2 + 2C_4L_1g_ms + 2C_4}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1 g_m}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: $\frac{L_1 g_m}{2C_4}$
K-HP: $\frac{L_4 g_m}{2C_1}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2+1},\ \infty,\ \infty,\ L_4 s + R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 g_m s^2 + C_4 L_1 R_4 g_m s + L_1 g_m}{2 C_1 C_4 L_1 s^2 + 2 C_4 L_1 g_m s + 2 C_4}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1 g_m}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: $\frac{L_1 g_m}{2 C_4}$
K-HP: $\frac{L_4 g_m}{2 C_1}$
K-BP: $\frac{R_4}{2}$
Qz: None
Wz: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ \infty,\ L_4 s + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 R_1 g_m s^2 + L_1 R_1 g_m}{2 C_1 C_4 L_1 R_1 s^2 + 2 C_4 R_1 + s\left(2 C_4 L_1 R_1 g_m + 2 C_4 L_1\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{C_1 \sqrt{L_1} R_1}$
K-LP: $\frac{L_1 g_m}{2 C_4}$
K-HP: $\frac{L_4 g_m}{2 C_1}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$

10.4 X-INVALID-WZ-4 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2+L_1 s+R_1},\ \infty,\ \infty,\ L_4 s + R_4 + \frac{1}{C_4 s},\ \infty,\ \infty\right)$

$$H(s) = \frac{C_4 L_1 L_4 R_1 g_m s^2 + C_4 L_1 R_1 R_4 g_m s + L_1 R_1 g_m}{2 C_1 C_4 L_1 R_1 s^2 + 2 C_4 R_1 + s\left(2 C_4 L_1 R_1 g_m + 2 C_4 L_1\right)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{C_1 \sqrt{L_1} R_1}$
K-LP: $\frac{L_1 g_m}{2 C_4}$
K-HP: $\frac{L_4 g_m}{2 C_1}$
K-BP: $\frac{R_1 R_4 g_m}{2 R_1 g_m + 2}$
Qz: None
Wz: $\frac{1}{\sqrt{C_4}\sqrt{L_4}}$

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